

Chapter 164 — Treatment of Locally Advanced Prostate Cancer — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

1. **≥10% of new diagnoses locally advanced** (T3, NX/N+, M0); no single modality suffices.
2. **Positive surgical margin** = highest relative risk (RR) of progression after radical prostatectomy (RP), above extracapsular extension (ECE), seminal vesicle invasion (SVI), node-positive (N+).
3. **Neoadjuvant androgen deprivation (NAD) before RP** → no gain in prostate-specific antigen (PSA) recurrence, disease-specific (DSS) or overall survival (OS).
4. **NAD + concurrent androgen deprivation (AD) with radiation therapy (RT)** → better control and cancer mortality (Radiation Therapy Oncology Group [RTOG] 86-10).
5. **High risk + RT → long AD**; National Comprehensive Cancer Network (NCCN): 18–36 mo external-beam (EBRT), 12–36 mo (brachytherapy boost).
6. **Intermediate risk + RT → 2–4 mo AD.**
7. **Early salvage RT preferred over adjuvant RT** for most (RADICALS-RT, GETUG-AFU 17, RAVES).
8. **Postoperative RT dose ≥64 Gy**; modern trials 64–66 Gy.
9. **Very high risk → RT + AD + 2 yr abiraterone** (STAMPEDE; NCCN 2023).
10. **Prostate-specific membrane antigen (PSMA) PET most accurate staging** (proPSMA 92% vs 65%); may replace CT + bone scan.

DEFINITION AND RISK ASSESSMENT

- **Cut-offs:** cT3a extracapsular; cT3b seminal vesicles; cT4 adjacent organs.
- **Most now cT1c** → PSA, Gleason, imaging define risk.
- **CAPRA (Cancer of the Prostate Risk Assessment) 0–10:** each +2 points doubles recurrence after RP.
- **Partin tables** → predicted pathological stage; 1990s, updated 2001/2013/2017.
- **Kattan RP nomogram** built on T1c–T3aNXM0 → **not for SVI or N+.**
- **D'Amico:** low / intermediate / high risk of biochemical failure.
- **Clinical-genomic groups (22-gene Decipher)** outperform NCCN and CAPRA.
- **Biochemical disease-free survival (BDFS)** = intermediate surrogate only.
- **Transrectal ultrasound (TRUS) understages;** no better than digital rectal examination (DRE).
- **MRI:** reserve for high risk (>3 cores, abnormal DRE, PSA >10); use PI-RADS v2.
- **⁶⁸Ga-PSMA-11 nodes:** specificity ≥95%, sensitivity ~40–66%.
- **Staging imaging (NCCN):** cT3–T4, PSA >20, grade group 4–5, primary pattern 5.
- **¹⁸F-fluciclovine** → biochemical recurrence; **¹⁸F-NaF** → bone.
- **CALGB 90203 genomics:** TMPRSS2-ERG, TP53, PTEN, FOXA1, SPOP; **no androgen receptor (AR) alteration.**

TRENDS AND NATURAL HISTORY

- **High risk now mostly Gleason-defined,** not clinical T3.
- **RP for CAPRA 6–10 rising;** commonest treatment by 2010–2013.
- **RP + extended pelvic lymph node dissection** only if completely excisable; else AD + RT.
- **Untreated grade 3** ≈ 10× disease-specific mortality (DSM) of grade 1.
- **Gleason 8–10 conservative:** 60–87% cancer death at 15 yr (pre-PSA); >20% (SEER registry).
- **Medical Research Council (MRC) delayed AD:** progression 10 mo, cancer death 48 mo (median).
- **Veterans Administration Cooperative Urological Research Group (VACURG) placebo, stage III:** 5-yr OS 58%.
- **Fit men >70 undertreated** (RT 40%, RP 10%) despite >10-yr life expectancy.

RADICAL PROSTATECTOMY

- **cT3 contemporary DSS:** 85–92% (5 yr), 79–82% (10 yr).
- **Clinical overstaging of cT3: 7–27%** → don't exclude on clinical stage.
- **Over a third** of high-risk RP specimens specimen-confined.
- **High-risk PSA-free survival after RP** ≈ 50% at 5–7 yr.
- **Up to half Gleason 8–10 / PSA >20** disease-free with RP alone.
- **5-yr clinical progression:** organ-confined (OC) 7% → ECE 18–35% → SVI 86% → N+ 95%.
- **5-yr PSA-free:** OC 95%, ECE 76%, SVI 37%, N+ 18%.
- **RR progression:** ECE 2.2–2.7, SVI 2.6, N+ 3.3, **margin 4.4.**
- **Biochemical recurrence (BCR) threshold 0.2** → 65–72% at 5 yr; **0.4** → 26%.
- **N+ poor predictors:** Gleason >8, >3 nodes.
- **Single positive node:** DSS >80%; most never need AD.
- **Minimally invasive = open** for high-risk outcomes.
- **Operate if:** completely excisable, lower-volume cT3; fit, long life expectancy.

Neoadjuvant:

- **NAD shrinks prostate 30–50%, PSA ~90%;** maximal by 2 mo.
- **Clinical downstaging 32–90% vs pathological 8–31%;** ~20% cT3 organ-confined.
- **cT3: margins not reduced;** cT1–T2 margins reduced, no PSA gain.
- **3 vs 8 mo NAD:** fewer margins (23→12%), same PSA recurrence.
- **CALGB 90203 (AD + docetaxel):** primary endpoint missed; metastasis-free survival (MFS), OS improved.
- **Neoadjuvant chemotherapy pT0:** 0–4%.

Adjuvant:

- **European Organisation for Research and Treatment of Cancer (EORTC) 22911:** BDFS 74% vs 53%; benefit only margin-positive (hazard ratio [HR] 0.38).
- **Southwest Oncology Group (SWOG) 8794:** MFS HR 0.71, OS HR 0.72; metastasis 16% → 7%.
- **ARO 96-02:** undetectable PSA, BDFS HR 0.53.
- **Adjuvant RT if:** extensive/multiple margins; very high risk (Gleason 8–10, pT3b).
- **pT3b favourable:** post-RP PSA <0.3 or positive margin.
- **Eastern Cooperative Oncology Group (ECOG) 7887:** N+ → immediate AD improves OS, DSS.
- **pN1: AD + RT > AD alone;** AD alone ↑ other-cause mortality (HR 3.05).

- **Adjuvant chemotherapy after RP** (S9921, VA CSP #553, SPCG-12): no benefit.
- **GETUG-AFU 16**: 3 mo AD + RT; progression-free survival HR 0.54.

RADIATION THERAPY

- **Dose**: ≥ 81 Gy high risk; 78 > 70 Gy, most if PSA >10.
- **RT alone, stage C**: OS 60–70% (5 yr), <50% (10 yr).
- **RTOG 86-10**: 4 mo AD, bulky ≥ 25 cm²; DSM 23% vs 36%; OS not significant.
- **TROG 96.01**: 6 mo NAD cuts all-cause mortality; 3 mo doesn't.
- **RTOG 9910**: 8 = 28 wk NAD (intermediate risk).
- **RTOG 85-31**: indefinite AD; OS gain Gleason 8–10 only.
- **EORTC 22863**: 3 yr AD, **only OS benefit** (79% vs 62%).
- **RTOG 92-02**: +24 mo AD; OS gain Gleason 8–10 (81% vs 71%).
- **18 = 36 mo AD** for OS; 18 mo better quality of life.
- **RTOG 94-13**: worst OS = whole pelvis (WP) + adjuvant AD.
- **ASCENDE-RT**: low-dose-rate (LDR) brachytherapy boost halves BCR vs EBRT boost (HR 2.04); OS equal.
- **Gleason 9–10**: EBRT + brachytherapy boost best (cancer mortality HR ~0.4).
- **RTOG 0521**: docetaxel + AD + RT → OS HR 0.69.
- **RTOG 9902**: no benefit; closed for thromboembolism.
- **GETUG 12**: docetaxel + estramustine → relapse-free HR 0.71.
- **STAMPEDE abiraterone M0**: MFS HR 0.53, OS HR 0.6.

FOCAL ABLATIVE THERAPY

- **Cryotherapy + high-intensity focused ultrasound (HIFU)** commonest; high-risk data limited.
- **Cryo vs RT (66 Gy)**: equal survival; more BCR (87% vs 53%).
- **Cryo at PSA 0.2–0.5 threshold** → >80% failure at 2 yr.
- **HIFU failure predictors**: $\geq$ cT2c (HR 2.34), no NAD (HR 2.19).

ANDROGEN DEPRIVATION AND ITS TIMING

- **VACURG**: diethylstilbestrol (DES) 1 mg ↑ OS; 5 mg → cardiovascular deaths.
- **Immediate oestrogen best**: <75 yr, Gleason 7–10.

- **MRC**: immediate AD ↓ cancer deaths (mainly M0).
- **SPCG-7 (Scandinavian Prostate Cancer Group)**: AD + RT → cancer mortality RR 0.44.
- **MRC PR07**: + RT → OS HR 0.77.
- **Deferred orchiectomy**: 42% never need AD; OS equal.
- **Bicalutamide 150 mg, localised**: death HR 1.23 → avoid.
- **Intermittent AD**: stop if PSA <4 by 24–32 wk; restart at 15.
- **Adjuvant AD (RTOG 85-31)**: no ↑ cardiovascular mortality.

DELAYED SEQUELAE

- **10-yr local recurrence after RT, cT3**: 24–74%.
- **After RT alone**: transurethral resection (TURP) ~1%, hydronephrosis 20%.
- **AD alone**: >half still need TURP; ureteric obstruction 25%.
- **Adjuvant RT**: worse bowel, urinary; **not erectile**.

CLASSIC TRAPS

1. **Trap**: NAD before RP helps like before RT — **Actually**: no survival benefit before RP.
2. **Trap**: clinical downstaging = pathological — **Actually**: only 8–31% pathological.
3. **Trap**: AD + RT improves OS — **Actually**: often disease-free survival only; OS in high-grade subsets.
4. **Trap**: ECE is worst RP finding — **Actually**: positive margin has highest RR.
5. **Trap**: Kattan RP nomogram for cT3b — **Actually**: not valid with SVI or N+.
6. **Trap**: Adjuvant RT for all pT3 — **Actually**: early salvage; very high risk underrepresented.
7. **Trap**: AD alone suffices for pN1 — **Actually**: AD + RT; AD alone ↑ other-cause death.
8. **Trap**: Whole-pelvis RT always better — **Actually**: RTOG 94-13 WP + adjuvant AD worst OS.
9. **Trap**: Undetectable PSA after chemo = pT0 — **Actually**: pT0 0–4%.
10. **Trap**: Bicalutamide 150 mg safe in localised — **Actually**: ↑ death (HR 1.23).